@henryleberre **Henry LE BERRE /ɑ̃ʁiləbɛʁ/** hberre3@gatech.edu

EDUCATION

- **Georgia Institute of Technology** (Atlanta, GA, USA) | Bachelor's in Science in **Computer Science** — July 2021 – Present
  GPA: 3.84 | Concentration: **Modeling & Simulation** and **Systems & Architecture**
- **Lycée Lavoisier** (Paris Vè, France) — September 2018 – July 2021

WORK EXPERIENCE

- **Space Exploration Technologies | Software Engineering Intern** — Los Angles, California, USA – Fall 2023
  Raptor Combustion Devices Engineering | Raptor Production Group

25% speed up of the Raptor Combustion CFD Code on production simulations through C++ CUDA GPU kernel optimizations.

- **Verkada | Software Engineering Intern** — San Mateo, California, USA – Summer 2023
  Device Platform Team | Alarms and Intrusion
- **Computational Physics Group | Student Assistant** — Atlanta, Georgia, USA – Summer 2022

PUBLICATIONS

- **[Preprint] Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow** — May 2023
  Submitted to **arXiv** and **Computer Methods in Applied Mechanics and Engineering**. — 10.48550/arXiv.2305.09163
- **Early Application Experiences on a Modern GPU-Accelerated Arm-based HPC Platform** — September 2022
  HPC Asia '23. International Workshop on Arm-based HPC: Practice and Experience (IWAHPCE). — 10.1145/3581576.3581621

TALKS, POSTERS, AND PRESENTATIONS

- **Scalable GPU Accelerated Simulation of Multiphase Compressible Flow** — November 15, 2022
  Supercomputing 2022 (SC'22), Dallas, Texas, USA — Poster | Paper
- **Exascale for humans: A development environment for MFC** — February 19, 2022
  Georgia Scientific Computing Symposium 2022 (GSCS 2022), Atlanta, Georgia, USA — Website

AWARDS

- **Verkadathon** (Verkada Hackathon) **Winner** ("Best Use of Verkada Products") — Summer 2023
- **President's Undergraduate Research Award** (**PURA**) **Travel** from the **Georgia Institute of Technology** — Fall 2022
- **Faculty Honors** and **Dean's List** from the **Georgia Institute of Technology**
- **National General Baccalaureate** (« Baccalauréat Général ») with **Highest Honors** (Mention « très bien ») — 2021
- **Highest Honors** (« Félicitations du conseil de classe ») for 9 consecutive trimesters — 2018 – 2021
- **Diplôme National du Brevet** with **Highest Honors** (Mention « très bien ») — 2018

RESEARCH AFFILIATIONS

- **Computational Physics Group (CPG) at Georgia Tech | Multi-Component Flow Code (MFC)** — November 2021 – Present

Resident GPU expert. Implementing reactive flow modeling (combustion). Offloaded and optimized the Fortran/MPI flow solver to efficiently run and scale simulations on leadership-class distributed systems of GPUs such as OLCF Summit, with PhD student Anand Radhakrishnan. Discovered and reported bugs in GNU and HPE Cray offloading compilers for Fortran, to support running on OLCF Frontier. Modernized the codebase, adding a test suite, CI, CMake support, and a custom build toolchain to enable simulation case-specific compile-time optimizations.

- **Space Systems Design Lab (SSDL) at Georgia Tech | GT-II Satellite** — November 2021 – Dec. 2022

Designed & implemented the the GT-II satellite's Over-The-Air update (OTA) protocols, on bare-metal, from ground uplink to satellite bootloader, hardware driver interfaces, downlink, and control logic.

LEADERSHIP, SERVICE, AND MISCELANIOUS

- **Student Volunteer** at **Supercomputing 2023 (SC'23)** | Denver, Colorado, USA — November 11 to 17 (2024)
- **Mentor** for team CEREAL at the **Georgia Tech Open Hackathon** (C++/MPI/Cuda/NVSHMEM) — January 2023
- **Co-Chair** of the **Georgia Tech French Club** — August 2022 – Present
- **Languages: English (Native | U.S. Citizen), French (Native | French Citizen), and German (High School Level)**